



HospitalOS

Hospital Operating System for Africa

Academy Certification Blueprint

The certification framework and architecture — not the full curriculum

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Scope of this document

This is a certification blueprint, not a training curriculum. It defines the framework each credential will be built against — objectives, audience, competency areas, proposed module structure, assessment approach, and how each credential connects to the next — so the full curriculum (lessons, exercises, quizzes, case studies, and the actual certification examinations) can be built later without redesigning the underlying architecture.

What this document is not. *It does not contain lesson content, exercises, quiz questions, or assessment items. Every certification described here is marked Soon in the HospitalOS Academy interface until that content exists.*

The pathway, at a glance

Tier	Certification	Code	Who it's for
1 — Foundation	HospitalOS Certified User	HCU	Every HospitalOS user, any role
2 — Professional	Certified Healthcare Digital Operations Professional	HCDOP	Operations, finance, and administrative staff
2 — Professional	Certified Healthcare Systems Administrator	HCSA	Hospital administrators and IT-facing Super Admins
3 — Advanced	Certified Clinical Digital Workflow Professional	HCDWP	Doctors, nurses, and diagnostic/pharmacy clinical staff
4 — Expert / Leadership	Certified Healthcare Digital Transformation Professional	HCDTP	Hospital leadership driving digital adoption

Deliberately five, not more. *Two Professional-tier credentials split by genuinely different daily work — operations/finance versus systems administration — rather than one broad, shallower certification. Everything above sits on HCU; HCDTP sits above both Professional tracks and HCDWP. No certification exists without a clear, distinct audience it actually serves.*

1. Tier 1 — Foundation — HospitalOS Certified User (HCU)

1.1 Overview and purpose

The single entry point to HospitalOS Academy. Every certified user, regardless of eventual specialisation, starts here — confirming they can navigate the system safely, understand why permission boundaries exist, and recognise the platform's core safety guarantees before touching any role-specific workflow.

1.2 Core objectives and learning outcomes

- Sign in, navigate the sidebar and search palette, and locate any screen the user's role permits without assistance.
- Correctly distinguish a genuine defect from expected role/permission behaviour before reporting either.
- Explain, in plain terms, what the alert system does and does not require of a user manually.
- Locate Help & documentation and the downloadable guides unprompted.

1.3 Intended audience and eligibility

Every individual granted a HospitalOS account, on day one, before any department-specific work begins. No prior digital-health experience assumed.

1.4 Recommended prerequisites

None. This is the only certification in the framework with no prerequisite — it is the prerequisite for everything else.

1.5 Certification level and progression

Tier 1 of 4. Mandatory foundation for HCDOP, HCSA, and HCDWP; indirectly required for HCDTP via those tracks.

1.6 Core competency areas

- Account and access fundamentals — sign-in, device recognition, credential security
- Navigation — the sidebar's area-based structure, the search palette
- The permission model — areas versus grants, and why a missing button usually isn't a bug
- The alert system — what raises an alert automatically, and why most clear themselves
- Getting help — in-app documentation, downloadable guides

1.7 Proposed curriculum structure

Four short modules, intentionally light — Foundation is meant to be completed in a single onboarding session, not a multi-week course.

Module	Focus
Module 1	Getting an account and signing in safely
Module 2	Navigating HospitalOS — sidebar, search, and your own role's view
Module 3	Understanding roles and permissions without assuming a bug
Module 4	The alert system, and where to get help

1.8 Expected practical skills and competencies

- Independently sign in and recover from a locked-out state through the correct channel
- Use Ctrl+K search effectively across screens, patients, and documentation
- Correctly identify a permission boundary versus a genuine fault, unprompted



1.9 Assessment and examination approach

A short knowledge-based assessment covering navigation, the permission model, and alert behaviour — no practical/scenario component at this tier, given the foundational scope.

1.10 Practical / project requirement

None at Foundation tier. Practical, scenario-based assessment begins at Professional tier and above.

1.11 Certification completion requirements

- Active HospitalOS account in good standing
- Pass the HCU knowledge assessment

1.12 The credential

HospitalOS Certified User (HCU) — a digital credential and certificate, recorded against the learner's account.

1.13 Validity and renewal

Not time-limited under the current framework — Foundation content is stable enough not to need periodic renewal. Revisited automatically if the underlying navigation or permission model changes materially.

1.14 Progression to the next level

HCU is the prerequisite for every Professional-tier track (HCDOP, HCSA) and for Advanced tier (HCDWP). A learner chooses their next certification based on their actual role, not a fixed sequence — Operations, Technology, and Clinical tracks are parallel, not sequential to each other.

2. Tier 2 — Professional — Certified Healthcare Digital Operations Professional (HCDOP)

2.1 Overview and purpose

For the staff who keep a hospital's non-clinical engine running — registration, billing, claims, and the operational departments that don't touch a patient's chart but are essential to the hospital functioning. Certifies genuine competence in HospitalOS's financial controls and administrative workflows, not just familiarity with the screens.

2.2 Core objectives and learning outcomes

- Operate a full cash session lifecycle correctly, including honest variance reporting.
- Correctly distinguish a refund from a chargeback and apply each appropriately.
- Move a claim through its full lifecycle, including co-payment collection, without violating separation of duties.
- Reconcile a bank statement import and resolve ambiguous matches responsibly.

2.3 Intended audience and eligibility

Cashiers, records officers, and staff in operational departments — scheduling, CSSD, biomedical engineering, facility management, fleet, and support services.

2.4 Recommended prerequisites

HCU (Foundation), completed and active.

2.5 Certification level and progression

Tier 2 of 4, parallel to HCSA — both sit directly above HCU, neither is a prerequisite for the other.

2.6 Core competency areas

- Patient registration and the ADT lifecycle
- Billing, cash sessions, and server-side balance validation
- Invoices, refunds, and chargebacks — the real distinctions between each
- Claims, insurers, and co-payment collection
- Bank reconciliation and payment-method-specific settlement behaviour
- Operational departments — scheduling, sterile supply, biomedical, facility, fleet

2.7 Proposed curriculum structure

Six modules, moving from patient-facing registration through to the financial controls that protect hospital revenue.

Module	Focus
Module 1	Registration and the patient record's starting point
Module 2	Billing and the five charge sources
Module 3	Cash sessions — opening, taking payment, closing honestly
Module 4	Invoices, refunds, and chargebacks
Module 5	Claims, insurers, and co-payments
Module 6	Bank reconciliation and the operational departments

2.8 Expected practical skills and competencies

- Run a complete, accurate cash session from open to close under realistic transaction volume



- Correctly apply a refund versus raising a chargeback given a real scenario
- Reconcile an imported bank statement, including handling at least one ambiguous match

2.9 Assessment and examination approach

Knowledge assessment covering all six modules, plus a scenario-based practical component simulating a full cashiering shift and a claims lifecycle end to end.

2.10 Practical / project requirement

A worked case study: reconcile a provided (simulated) bank statement against a set of pending Bank Transfer payments, correctly resolving both automatic and manual matches.

2.11 Certification completion requirements

- Active HCU certification
- Pass the HCDOP knowledge assessment
- Complete the practical reconciliation case study to the required standard

2.12 The credential

HospitalOS Certified Healthcare Digital Operations Professional (HCDOP) — digital credential, certificate, and a distinct badge from HCSA reflecting the operations specialisation.

2.13 Validity and renewal

Renewal recommended every 24 months, or immediately upon a material change to billing, claims, or reconciliation workflows — whichever comes first.

2.14 Progression to the next level

Feeds into HCDTP (Expert/Leadership) alongside HCSA and HCDWP — HCDTP requires demonstrated competence across operational and at least one other track, not HCDOP alone.

3. Tier 2 — Professional — Certified Healthcare Systems Administrator (HCSA)

3.1 Overview and purpose

For the people responsible for a hospital's own HospitalOS configuration — staff, roles, pricing, branding, and the audit trail. Certifies the judgement to make administrative changes deliberately, understanding exactly what each change affects and why certain protections (like the last-admin safeguard) exist.

3.2 Core objectives and learning outcomes

- Create and assign staff accounts with the correct role, understanding the weight of each grant.
- Set and override pricing correctly, understanding exactly what does and doesn't get rewritten.
- Read the audit trail confidently to answer a real “who did what, when” question.
- Explain the area/grant permission model precisely enough to diagnose a colleague's access issue.

3.3 Intended audience and eligibility

Hospital administrators, IT-facing staff, and any Super Admin account holder responsible for their hospital's own HospitalOS configuration.

3.4 Recommended prerequisites

HCU (Foundation), completed and active.

3.5 Certification level and progression

Tier 2 of 4, parallel to HCDOP — both sit directly above HCU, neither is a prerequisite for the other.

3.6 Core competency areas

- User and role management, including the Super Admin trust boundary
- Pricing configuration and its immediate, hospital-wide effect
- Branding and hospital identity settings
- Document and template management
- The audit trail — append-only, hash-chained, and how to actually use it
- RBAC — the area/grant model, explained precisely enough to teach it

3.7 Proposed curriculum structure

Five modules, oriented around configuration decisions with real, hospital-wide consequences.

Module	Focus
Module 1	Users, roles, and the weight of Super Admin
Module 2	Pricing — setting it, and what does and doesn't change retroactively
Module 3	Branding, identity, and documents & templates
Module 4	The audit trail — reading it, trusting it correctly
Module 5	RBAC in depth — areas, grants, and diagnosing access issues

3.8 Expected practical skills and competencies

- Correctly diagnose and resolve a real staff-member access complaint using the role matrix, not guesswork
- Set a price override and correctly predict its effect on past versus future charges



— Answer a “who did this, and when” question directly from the audit trail

3.9 Assessment and examination approach

Knowledge assessment across all five modules, plus a scenario-based practical component built around real support-style situations (a locked-out user, a disputed price, an audit query).

3.10 Practical / project requirement

A worked access-diagnosis case study: given a described complaint and the relevant role's actual permission list, determine whether it's a bug or expected behaviour, and propose the correct resolution.

3.11 Certification completion requirements

- Active HCU certification
- Pass the HCSA knowledge assessment
- Complete the access-diagnosis case study to the required standard

3.12 The credential

HospitalOS Certified Healthcare Systems Administrator (HCSA) — digital credential, certificate, and a distinct badge from HCDOP reflecting the systems-administration specialisation.

3.13 Validity and renewal

Renewal recommended every 24 months, or immediately upon a material change to the RBAC model, pricing engine, or audit system — whichever comes first.

3.14 Progression to the next level

Feeds into HCDTP (Expert/Leadership) alongside HCDOP and HCDWP — HCDTP requires demonstrated competence across systems administration and at least one other track, not HCSA alone.

4. Tier 3 — Advanced — Certified Clinical Digital Workflow Professional (HCDWP)

4.1 Overview and purpose

For clinical and diagnostic staff whose daily work is direct patient care through HospitalOS — the medical record, ward and critical-care workflows, laboratory and imaging, and pharmacy. Sits at Advanced tier, above the Professional tier, in recognition that clinical workflow competence carries direct patient-safety weight the administrative tracks don't.

4.2 Core objectives and learning outcomes

- File and correctly amend clinical documentation within HospitalOS's immutable-record model.
- Operate ward, ICU, and specialist-unit workflows without triggering avoidable alerts through delay or error.
- Move a diagnostic order through its full lifecycle, respecting the deliberate result/verification separation.
- Dispense safely, understanding both hard-blocked safety checks and why they cannot be overridden.

4.3 Intended audience and eligibility

Doctors, nurses, lab scientists, radiographers, and pharmacists — any role whose grants include Patient care, Diagnostics, or Pharmacy.

4.4 Recommended prerequisites

HCU (Foundation), completed and active. HCDWP does not require HCDOP or HCSA — clinical competence stands independently of administrative competence.

4.5 Certification level and progression

Tier 3 of 4 — the only Advanced-tier certification in the current framework, positioned above Professional tier given its direct patient-safety weight.

4.6 Core competency areas

- The medical record — SOAP notes, the amendment model, and why nothing is ever edited in place
- Registration, admission, and the ward/bed board's real-time integrity
- ICU/HDU vitals and the automatic critical-alert threshold
- The full laboratory order lifecycle, including the result/verification separation
- Imaging, radiology, and when a study becomes billable
- Blood bank crossmatching and its proven concurrency safety
- Pharmacy dispensing — the allergy and stock hard-blocks

4.7 Proposed curriculum structure

Seven modules — the largest track in the framework, reflecting the breadth of clinical workflow HospitalOS actually covers.

Module	Focus
Module 1	The medical record and the amendment principle
Module 2	Registration, admission, and the ward board
Module 3	ICU, HDU, and critical alerting
Module 4	The laboratory order lifecycle
Module 5	Imaging and radiology

Module	Focus
Module 6	Blood bank and crossmatching
Module 7	Pharmacy dispensing and its hard safety blocks

4.8 Expected practical skills and competencies

- File a clinical note and correctly perform an amendment rather than attempt an edit
- Recognise and respond correctly to an automatically-raised critical alert
- Trace a diagnostic order through every lifecycle stage, explaining why entry and verification are separated
- Explain, precisely, why the allergy and stock checks cannot be overridden even under pressure

4.9 Assessment and examination approach

Knowledge assessment across all seven modules, plus a scenario-based practical examination built around a realistic patient journey spanning multiple clinical touchpoints.

4.10 Practical / project requirement

A worked patient-journey case study spanning registration through at least one diagnostic order and one dispense, requiring the candidate to correctly navigate every safety check along the way.

4.11 Certification completion requirements

- Active HCU certification
- Pass the HCDWP knowledge assessment
- Complete the patient-journey case study to the required standard

4.12 The credential

HospitalOS Certified Clinical Digital Workflow Professional (HCDWP) — digital credential and certificate, the framework's Advanced-tier credential.

4.13 Validity and renewal

Renewal recommended every 18 months, shorter than the Professional tracks, given the direct patient-safety relevance of clinical workflow changes — or immediately upon a material change to any clinical safety check.

4.14 Progression to the next level

Feeds into HCDTP (Expert/Leadership) alongside HCDOP and HCSA — HCDTP requires demonstrated competence across clinical workflow and at least one administrative track, not HCDWP alone.

5. Tier 4 — Expert / Leadership — Certified Healthcare Digital Transformation Professional (HCDTP)

5.1 Overview and purpose

The framework's capstone. Not a deeper version of any single track, but a cross-functional credential for those responsible for a hospital's overall digital adoption — understanding clinical, operational, and administrative workflows well enough to lead change, make platform-wide decisions, and connect HospitalOS's technical guarantees to real hospital governance and strategy.

5.2 Core objectives and learning outcomes

- Explain, to hospital leadership and to AgoroX, how HospitalOS's core guarantees (auditability, separation of duties, safety hard-blocks) map to real governance and risk obligations.
- Evaluate a proposed workflow or policy change for its effect across clinical, operational, and administrative tracks simultaneously.
- Read and act on cross-cutting operational intelligence — analytics, forecasting, and compliance posture together.
- Lead a hospital's own staff certification programme across the Foundation and Professional tiers.

5.3 Intended audience and eligibility

Hospital leadership, medical directors, and senior administrators accountable for a hospital's digital operations as a whole — not a specific department.

5.4 Recommended prerequisites

HCU (Foundation), plus demonstrated competence across HCDWP and at least one of HCDOP or HCSA — reflecting genuine cross-functional grounding, not a single specialisation.

5.5 Certification level and progression

Tier 4 of 4 — the top of the framework. No further tier above HCDTP under the current architecture.

5.6 Core competency areas

- Cross-functional fluency — clinical, operational, and administrative workflows understood together, not separately
- Governance and audit literacy — connecting HospitalOS's technical guarantees to real accountability
- Intelligence and forecasting — reading analytics as a planning signal, not a precise prediction
- Compliance posture — licenses, accreditation, incident management, and policy currency, viewed as one picture
- Change leadership — evaluating and sequencing a digital workflow change across an entire hospital

5.7 Proposed curriculum structure

Four modules — deliberately fewer than the tracks beneath it, since HCDTP builds on competence already certified elsewhere rather than teaching workflow mechanics again.

Module	Focus
Module 1	Governance, audit, and risk — what HospitalOS's guarantees actually mean
Module 2	Reading the hospital as one system — intelligence, forecasting, and compliance together
Module 3	Leading a change — evaluating a workflow decision across every track
Module 4	Leading certification — growing Foundation and Professional competence hospital-wide

5.8 Expected practical skills and competencies

- Evaluate a proposed operational change and correctly identify its clinical, financial, and compliance consequences before it's made
- Present a hospital's own audit and compliance posture credibly to an external party
- Design a realistic staff certification rollout plan across Foundation and Professional tiers for a real hospital

5.9 Assessment and examination approach

A capstone assessment combining a cross-functional written examination with a leadership-level case study — no separate knowledge/practical split, reflecting that judgement and knowledge are being assessed together at this tier.

5.10 Practical / project requirement

A capstone project: a written digital-transformation plan for a representative hospital, addressing at least one clinical, one operational, and one governance consideration, evaluated for coherence across all three.

5.11 Certification completion requirements

- Active HCU certification
- Active HCDWP certification
- Active HCDOP or HCSA certification (at least one)
- Pass the HCDTP capstone assessment
- Submit and pass the capstone transformation project

5.12 The credential

HospitalOS Certified Healthcare Digital Transformation Professional (HCDTP) — the framework's highest credential, digital and physical certificate, with a distinct leadership-tier badge.

5.13 Validity and renewal

Renewal every 24 months, alongside a standing expectation of continued active status on all prerequisite certifications — HCDTP lapses if its underlying foundation does.

5.14 Progression to the next level

No further tier under the current framework. HCDTP holders are the natural pool for leading a hospital's own internal certification rollout (Module 4), extending the Academy's reach without requiring AgoroX involvement in every hospital's staff training.

Building for later, without redesigning now

Every certification above is defined by its objectives, competencies, and proposed module list — deliberately, so the actual lesson content, exercises, quizzes, case studies, and examination items can be authored later against this exact structure without renumbering, resequencing, or redefining what each credential is for. The module tables in this document are the scaffolding the real curriculum gets built onto, not a placeholder to be discarded.

What stays locked in the app until then. Every certification described here shows as *Soon* in the HospitalOS Academy interface. Nothing here is offered, sold, or awarded until the actual curriculum and assessments exist — this document is the blueprint the build works from, not an early or partial release of the programme itself.